Modify Registry

Adversaries may interact with the Windows Registry as part of a variety of other techniques to aid in defense evasion, persistence, and execution.

Access to specific areas of the Registry depends on account permissions, with some keys requiring administrator-level access. The built-in Windows command-line utility Reg may be used for local or remote Registry modification.[1] Other tools, such as remote access tools, may also contain functionality to interact with the Registry through the Windows API.

The Registry may be modified in order to hide configuration information or malicious payloads via Obfuscated Files or Information.[2][3][4][5] The Registry may also be modified to impair defenses, such as by enabling macros for all Microsoft Office products, allowing privilege escalation without alerting the user, increasing the maximum number of allowed outbound requests, and/or modifying systems to store plaintext credentials in memory.[6][2]

The Registry of a remote system may be modified to aid in execution of files as part of lateral movement. It requires the remote Registry service to be running on the target system.[7] Often Valid Accounts are required, along with access to the remote system's SMB/Windows Admin Shares for RPC communication.

Finally, Registry modifications may also include actions to hide keys, such as prepending key names with a null character, which will cause an error and/or be ignored when read via Reg or other utilities using the Win32 API.[8] Adversaries may abuse these pseudo-hidden keys to conceal payloads/commands used to maintain persistence.[9][10]

ID: T1112

Sub-techniques: No sub-techniques

ⓘ Tactics: Defense Impairment, Persistence

ⓘ Platforms: Windows

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Procedure Examples

ID	Name	Description
C0028	2015 Ukraine Electric Power Attack	During the 2015 Ukraine Electric Power Attack, Sandworm Team modified in-registry Internet settings to lower internet security before launching `rundll32.exe`, which in-turn launches the malware and communicates with C2 servers over the Internet. [11].
S0677	AADInternals	AADInternals can modify registry keys as part of setting a new pass-through authentication agent.[12]
S0045	ADVSTORESHELL	ADVSTORESHELL is capable of setting and deleting Registry values.[13]
S0331	Agent Tesla	Agent Tesla can achieve persistence by modifying Registry key entries.[14]
S1025	Amadey	Amadey has overwritten registry keys for persistence.[15]
G0073	APT19	APT19 uses a Port 22 malware variant to modify several Registry keys.[16]
G0050	APT32	APT32's backdoor has modified the Windows Registry to store the backdoor's configuration. [17]
G0082	APT38	APT38 uses a tool called CLEANTOAD that has the capability to modify Registry keys.[18]
G0096	APT41	APT41 used a malware variant called GOODLUCK to modify the registry in order to steal credentials.[19][20]
G1044	APT42	APT42 has modified Registry keys to maintain persistence.[21]
G0143	Aquatic Panda	Aquatic Panda modified the victim registry to enable the `RestrictedAdmin` mode feature, allowing for pass the hash behaviors to function via RDP.[22]
S0438	Attor	Attor's dispatcher can modify the Run registry key.[23]
S0640	Avaddon	Avaddon modifies several registry keys for persistence and UAC bypass.[24]
S0031	BACKSPACE	BACKSPACE is capable of deleting Registry keys, sub-keys, and values on a victim system.[25]
S0245	BADCALL	BADCALL modifies the firewall Registry key `SYSTEM\CurrentControlSet\Services\SharedAccess\Parameters\FirewallPolicy\StandardProfileGloballyOpenPorts\List`.[26]
S0239	Bankshot	Bankshot writes data into the Registry key `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Pniumj`.[27]
S0268	Bisonal	Bisonal has deleted Registry keys to clean up its prior activity.[28]
S0570	BitPaymer	BitPaymer can set values in the Registry to help in execution.[29]
S1070	Black Basta	Black Basta has modified the Registry to enable itself to run in safe mode, to change the icons and file extensions for encrypted files, and to add the malware path for persistence.[30][31][32][33][34][35]
G1043	BlackByte	BlackByte performed Registry modifications to escalate privileges and disable security tools.[36][37]
S1181	BlackByte 2.0 Ransomware	BlackByte 2.0 Ransomware modifies the victim Registry to allow for elevated execution.[38]
S1180	BlackByte Ransomware	BlackByte Ransomware modifies the victim Registry to prevent system recovery.[39]

ID	Name	Description
S0348	Cardinal RAT	Cardinal RAT sets `HKCU\Software\Microsoft\Windows NT\CurrentVersion\Windows\Load` to point to its executable.[42]
S0261	Catchamas	Catchamas creates three Registry keys to establish persistence by adding a Windows Service.[43]
S0572	Caterpillar WebShell	Caterpillar WebShell has a command to modify a Registry key.[44]
S0631	Chaes	Chaes can modify Registry values to stored information and establish persistence.[45]
S0674	CharmPower	CharmPower can remove persistence-related artifacts from the Registry.[46]
S1149	CHIMNEYSWEEP	CHIMNEYSWEEP can use the Windows Registry Environment key to change the `%windir%` variable to point to `c:\Windows` to enable payload execution.[47]
S0023	CHOPSTICK	CHOPSTICK may modify Registry keys to store RC4 encrypted configuration information.[48]
S0660	Clambling	Clambling can set and delete Registry keys.[49]
S0611	Clop	Clop can make modifications to Registry keys.[50]
S0154	Cobalt Strike	Cobalt Strike can modify Registry values within `HKEY_CURRENT_USER\Software\Microsoft\Office\\Excel\Security\AccessVBOM\` to enable the execution of additional code.[51]
S0126	ComRAT	ComRAT has modified Registry values to store encrypted orchestrator code and payloads.[52][53]
S0608	Conficker	Conficker adds keys to the Registry at `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services` and various other Registry locations.[54][55]
S0488	CrackMapExec	CrackMapExec can create a registry key using wdigest.[56]
S0115	Crimson	Crimson can set a Registry key to determine how long it has been installed and possibly to indicate the version number.[57]
S0527	CSPY Downloader	CSPY Downloader can write to the Registry under the `%windir%` variable to execute tasks.[58]
S0334	DarkComet	DarkComet adds a Registry value for its installation routine to the Registry Key `HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\System Enable LUA="0"` and `HKEY_CURRENT_USER\Software\DC3_FEXEC`.[59][60]
S1066	DarkTortilla	DarkTortilla has modified registry keys for persistence.[61]
S0673	DarkWatchman	DarkWatchman can modify Registry values to store configuration strings, keylogger, and output of components.[62]
S1033	DCSrv	DCSrv has created Registry keys for persistence.[63]
G0035	Dragonfly	Dragonfly has modified the Registry to perform multiple techniques through the use of Reg.[64]
G1006	Earth Lusca	Earth Lusca modified the registry using the command `reg add "HKEY_CURRENT_USER\Environment" /v UserInitMprLogonScript /t REG_SZ /d "[file path]"` for persistence.[65]
S1247	Embargo	Embargo has modified and deleted Registry keys to add services, and to disable Security Solutions such as Windows Defender.[66]

ID	Name	Description
S0569	Explosive	Explosive has a function to write itself to Registry values.[70]
S0267	FELIXROOT	FELIXROOT deletes the Registry key `HKCU\Software\Classes\Applications\rundll32.exe\shell\open`.[71]
S0679	Ferocious	Ferocious has the ability to add a Class ID in the current user Registry hive to enable persistence mechanisms.[72]
G0061	FIN8	FIN8 has deleted Registry keys during post compromise cleanup activities.[73]
G0047	Gamaredon Group	Gamaredon Group has removed security settings for VBA macro execution by changing registry values `HKCU\Software\Microsoft\Office\<version>\<product>\Security\VBAWarnings` and `HKCU\Software\Microsoft\Office\<version>\<product>\Security\AccessVBOM`.[74][75][76] Gamaredon Group has also modified Registry keys to hide folders and system files and to add the C2 address under `HKEY_CURRENT_USER\Console\WindowsUpdate`. [77]
S0666	Gelsemium	Gelsemium can modify the Registry to store its components.[78]
S0032	gh0st RAT	gh0st RAT has altered the InstallTime subkey.[79]
G0078	Gorgon Group	Gorgon Group malware can deactivate security mechanisms in Microsoft Office by editing several keys and values under `HKCU\Software\Microsoft\Office\`.[80]
S0531	Grandoreiro	Grandoreiro can modify the Registry to store its configuration at `HKCU\Software\` under frequently changing names including `%USERNAME%` and `ToolTech-RM`.[81]
S0342	GreyEnergy	GreyEnergy modifies conditions in the Registry and adds keys.[82]
S0697	HermeticWiper	HermeticWiper has the ability to modify Registry keys to disable crash dumps, colors for compressed files, and pop-up information about folders and desktop items.[83][84][85]
S9023	HiddenFace	HiddenFace can store its configuration file in the Registry.[86]
S1230	HIUPAN	HIUPAN has modified registry keys to ensure hidden files and extensions are not visible through the modification of `HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Advanced`.[87][88]
S0376	HOPLIGHT	HOPLIGHT has modified Managed Object Format (MOF) files within the Registry to run specific commands and create persistence on the system.[89]
S0203	Hydraq	Hydraq creates a Registry subkey to register its created service, and can also uninstall itself later by deleting this value. Hydraq's backdoor also enables remote attackers to modify and delete subkeys.[90][91]
S0537	HyperStack	HyperStack can add the name of its communication pipe to `HKLM\SYSTEM\CurrentControlSet\Services\lanmanserver\parameters\NullSessionPipes`.[92]
G0119	Indrik Spider	Indrik Spider has modified registry keys to prepare for ransomware execution and to disable common administrative utilities.[93]
S0260	InvisiMole	InvisiMole has a command to create, set, copy, or delete a specified Registry key or value.[94][95]
S1132	IPsec Helper	IPsec Helper can make arbitrary changes to registry keys based on provided input.[96]
S1190	Kapeka	Kapeka writes persistent configuration information to the victim host registry.[97]
S0271	KEYMARBLE	KEYMARBLE has a command to create Registry entries for storing data under `HKEY_CURRENT_USER\SOFTWARE\Microsoft\WABE\DataPath`.[98]

ID	Name	Description
S0356	KONNI	KONNI has modified registry keys of ComSysApp, Svchost, and xmlProv on the machine to gain persistence.[105][106]
S1199	LockBit 2.0	LockBit 2.0 can create Registry keys to bypass UAC and for persistence.[107]
S1202	LockBit 3.0	LockBit 3.0 can change the Registry values for Group Policy refresh time, to disable SmartScreen, and to disable Windows Defender.[108][109]
S0397	LoJax	LoJax has modified the Registry key `'HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Session Manager\BootExecute'` from `'autocheck autochk *'` to `'autocheck autoche *'`.[110]
S0447	Lokibot	Lokibot has modified the Registry as part of its UAC bypass process.[111]
G0030	Lotus Blossom	Lotus Blossom has installed tools such as Sagerunex by writing them to the Windows registry.[112]
G1014	LuminousMoth	LuminousMoth has used malware that adds Registry keys for persistence.[113][114]
S1060	Mafalda	Mafalda can manipulate the system registry on a compromised host.[115]
G0059	Magic Hound	Magic Hound has modified Registry settings for security tools.[116]
G1051	Medusa Group	Medusa Group has modified Registry keys to elevate privileges, maintain persistence and allow remote access.[117]
S0576	MegaCortex	MegaCortex has added entries to the Registry for ransom contact information.[118]
S1059	metaMain	metaMain can write the process ID of a target process into the `HKEY_LOCAL_MACHINE\SOFTWARE\DDE\tpid` Registry value as part of its reflective loading activity.[115]
S0455	Metamorfo	Metamorfo has written process names to the Registry, disabled IE browser features, deleted Registry keys, and changed the ExtendedUIHoverTime key.[119][120][121][122]
S1047	Mori	Mori can write data to `HKLM\Software\NFC\IPA` and `HKLM\Software\NFC\` and delete Registry values.[123][124]
S0256	Mosquito	Mosquito can modify Registry keys under `HKCU\Software\Microsoft[dllname]` to store configuration values. Mosquito also modifies Registry keys under `HKCR\CLSID...\InprocServer32` with a path to the launcher.[125]
S9032	MuddyViper	MuddyViper has the ability to clear the Registry values in the Windows Startup folder that were previously set for persistence.[126]
S0205	Naid	Naid creates Registry entries that store information about a created service and point to a malicious DLL dropped to disk.[127]
S0336	NanoCore	NanoCore has the capability to edit the Registry.[128][129]
S0691	Neoichor	Neoichor has the ability to configure browser settings by modifying Registry entries under `HKEY_CURRENT_USER\Software\Microsoft\Internet Explorer`.[130]
S0210	Nerex	Nerex creates a Registry subkey that registers a new service.[131]
S0457	Netwalker	Netwalker can add the following registry entry: `HKEY_CURRENT_USER\SOFTWARE{8 random characters}`.[132]
S0198	NETWIRE	NETWIRE can modify the Registry to store its configuration information.[133]

ID	Name	Description
S9025	NOOPLDR	NOOPLDR can store its payload in the Registry using a random hex string in `HKCU\SOFTWARE\Microsoft\COM3`.[138]
S1131	NPPSPY	NPPSPY modifies the Registry to record the malicious listener for output from the Winlogon process.[139]
G0049	OilRig	OilRig has used reg.exe to modify system configuration.[140][141]
C0006	Operation Honeybee	During Operation Honeybee, the threat actors used batch files that modified registry keys.[142]
C0014	Operation Wocao	During Operation Wocao, the threat actors enabled Wdigest by changing the `HKLM\SYSTEM\\ControlSet001\\Control\\SecurityProviders\\WDigest` registry value from 0 (disabled) to 1 (enabled).[143]
S0229	Orz	Orz can perform Registry operations.[144]
S0664	Pandora	Pandora can write an encrypted token to the Registry to enable processing of remote commands.[145]
G0040	Patchwork	A Patchwork payload deletes Resiliency Registry keys created by Microsoft Office applications in an apparent effort to trick users into thinking there were no issues during application runs.[146]
S1050	PcShare	PcShare can delete its persistence mechanisms from the registry.[147]
S0158	PHOREAL	PHOREAL is capable of manipulating the Registry.[148]
S0517	Pillowmint	Pillowmint has modified the Registry key `HKLM\SOFTWARE\Microsoft\DRM` to store a malicious payload.[149]
S0501	PipeMon	PipeMon has modified the Registry to store its encrypted payload.[150]
S0254	PLAINTEE	PLAINTEE uses `reg add` to add a Registry Run key for persistence.[151]
S0013	PlugX	PlugX has a module to create, delete, or modify Registry keys.[152][153][154]
S0428	PoetRAT	PoetRAT has made registry modifications to alter its behavior upon execution.[155]
S0012	PoisonIvy	PoisonIvy creates a Registry subkey that registers a new system device.[156]
S0518	PolyglotDuke	PolyglotDuke can write encrypted JSON configuration files to the Registry.[157]
S0441	PowerShower	PowerShower has added a registry key so future powershell.exe instances are spawned off-screen by default, and has removed all registry entries that are left behind during the dropper process.[158]
S1058	Prestige	Prestige has the ability to register new registry keys for a new extension handler via `HKCR\.enc` and `HKCR\enc\shell\open\command`.[159]
S0583	Pysa	Pysa has modified the registry key "SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System" and added the ransom note.[160]
S0650	QakBot	QakBot can modify the Registry to store its configuration information in a randomly named subkey under `HKCU\Software\Microsoft`.[161][162]
S1242	Qilin	Qilin can make Registry modifications to share networked drives between elevated and non-elevated processes and to increase the number of outstanding network requests per client.[163][164] Qilin can also modify `HKEY_CURRENT_USER\Control Panel\Desktop\Wallpaper` to enable posting of ransom messages.[165]

ID	Name	Description
S0075	Reg	Reg may be used to interact with and modify the Windows Registry of a local or remote system at the command-line interface.[1]
S0511	RegDuke	RegDuke can create seemingly legitimate Registry key to store its encryption key.[157]
S0019	Regin	Regin appears to have functionality to modify remote Registry information.[170]
S0332	Remcos	Remcos has full control of the Registry, including the ability to modify it.[171][172]
S0496	REvil	REvil can modify the Registry to save encryption parameters and system information.[173][174][175][176][177]
S0240	ROKRAT	ROKRAT can modify the `HKEY_CURRENT_USER\Software\Microsoft\Office\` registry key so it can bypass the VB object model (VBOM) on a compromised host.[178]
S0090	Rover	Rover has functionality to remove Registry Run key persistence as a cleanup procedure.[179]
S0148	RTM	RTM can delete all Registry entries created during its execution.[180]
G1031	Saint Bear	Saint Bear will leverage malicious Windows batch scripts to modify registry values associated with Windows Defender functionality.[181]
S1099	Samurai	The Samurai loader component can create multiple Registry keys to force the svchost.exe process to load the final backdoor.[182]
S0596	ShadowPad	ShadowPad can modify the Registry to store and maintain a configuration block and virtual file system.[183][65]
S0140	Shamoon	Once Shamoon has access to a network share, it enables the RemoteRegistry service on the target system. It will then connect to the system with RegConnectRegistryW and modify the Registry to disable UAC remote restrictions by setting `SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\LocalAccountTokenFilterPolicy` to 1.[184][185][186]
C0058	SharePoint ToolShell Exploitation	During SharePoint ToolShell Exploitation, threat actors, including Storm-2603, disabled security services via Registry modifications.[187]
S0444	ShimRat	ShimRat has registered two registry keys for shim databases.[188]
S1178	ShrinkLocker	ShrinkLocker modifies various registry keys associated with system logon and BitLocker functionality to effectively lock-out users following disk encryption.[189][190]
S0589	Sibot	Sibot has modified the Registry to install a second-stage script in the `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\sibot`.[191]
G0091	Silence	Silence can create, delete, or modify a specified Registry key or value.[192]
S0692	SILENTTRINITY	SILENTTRINITY can modify registry keys, including to enable or disable Remote Desktop Protocol (RDP).[193]
S0533	SLOTHFULMEDIA	SLOTHFULMEDIA can add, modify, and/or delete registry keys. It has changed the proxy configuration of a victim system by modifying the `HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap` registry.[194]
S0649	SMOKEDHAM	SMOKEDHAM has modified registry keys for persistence, to enable credential caching for credential access, and to facilitate lateral movement via RDP.[195]

ID	Name	Description
S0559	SUNBURST	SUNBURST had commands that allow an attacker to write or delete registry keys, and was observed stopping services by setting their `HKLM\SYSTEM\CurrentControlSet\services\[service_name]\Start` registry entries to value 4.[198][199] It also deleted previously-created Image File Execution Options (IFEO) Debugger registry values and registry keys related to HTTP proxy to clean up traces of its activity.[200]
S0242	SynAck	SynAck can manipulate Registry keys.[201]
S0663	SysUpdate	SysUpdate can write its configuration file to `Software\Classes\scConfig` in either `HKEY_LOCAL_MACHINE` or `HKEY_CURRENT_USER`.[145]
G0092	TA505	TA505 has used malware to disable Windows Defender through modification of the Registry.[202]
S0011	Taidoor	Taidoor has the ability to modify the Registry on compromised hosts using `RegDeleteValueA` and `RegCreateKeyExA`.[203]
S0467	TajMahal	TajMahal can set the `KeepPrintedJobs` attribute for configured printers in `SOFTWARE\Microsoft\Windows NT\CurrentVersion\Print\Printers` to enable document stealing.[204]
S1011	Tarrask	Tarrask is able to delete the Security Descriptor (`SD`) registry subkey in order to "hide" scheduled tasks.[205]
S0560	TEARDROP	TEARDROP modified the Registry to create a Windows service for itself on a compromised host.[206]
G0027	Threat Group-3390	A Threat Group-3390 tool has created new Registry keys under `HKEY_CURRENT_USER\Software\Classes\` and `HKLM\SYSTEM\CurrentControlSet\services`.[207][145]
S0665	ThreatNeedle	ThreatNeedle can modify the Registry to save its configuration data as the following RC4-encrypted Registry key: `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\GameCon`.[208]
S0668	TinyTurla	TinyTurla can set its configuration parameters in the Registry.[209]
S1201	TRANSLATEXT	TRANSLATEXT has modified the following registry key to install itself as the value, granting permission to install specified extensions: `HKCU\Software\Policies\Google\Chrome\ExtensionInstallForcelist`.[210]
S0266	TrickBot	TrickBot can modify registry entries.[211]
G0010	Turla	Turla has modified Registry values to store payloads.[212][213]
S0263	TYPEFRAME	TYPEFRAME can install encrypted configuration data under the Registry key `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\ShellCompatibility\Applications\laxhost.dll` and `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\PrintConfigs`.[214]
S0022	Uroburos	Uroburos can store configuration information in the Registry including the initialization vector and AES key needed to find and decrypt other Uroburos components.[215]
S0386	Ursnif	Ursnif has used Registry modifications as part of its installation routine.[216][217]
S0476	Valak	Valak has the ability to modify the Registry key `HKCU\Software\ApplicationContainer\Appsw64` to store information regarding the C2 server and downloads.[218][219][220]
S0180	Volgmer	Volgmer modifies the Registry to store an encoded configuration file in `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\WMI\Security`.[221][222]
G1017	Volt Typhoon	Volt Typhoon has used `netsh` to create a PortProxy Registry modification on a compromised server running the Paessler Router Traffic Grapher (PRTG).[223]

ID	Name	Description
G0102	Wizard Spider	Wizard Spider has modified the Registry key `HKLM\System\CurrentControlSet\Control\SecurityProviders\WDigest` by setting the `UseLogonCredential` registry value to `1` in order to force credentials to be stored in clear text in memory. Wizard Spider has also modified the WDigest registry key to allow plaintext credentials to be cached in memory.[228][229]
S0330	Zeus Panda	Zeus Panda modifies several Registry keys under `HKCU\Software\Microsoft\Internet Explorer\PhishingFilter\` to disable phishing filters.[230]
S0350	zwShell	zwShell can modify the Registry.[134]
S0412	ZxShell	ZxShell can create Registry entries to enable services to run.[231]

Mitigations

ID	Mitigation	Description
M1024	Restrict Registry Permissions	Ensure proper permissions are set for Registry hives to prevent users from modifying keys for system components that may lead to privilege escalation.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0280	Behavior-Based Registry Modification Detection on Windows	AN0781	Behavior chain involving abnormal registry modifications via CLI, PowerShell, WMI, or direct API calls, especially targeting persistence, privilege escalation, or defense evasion keys, potentially followed by service restart or process execution. Such as editing Notify/Userinit/Startup keys, or disabling SafeDllSearchMode.

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